

Real-Attack Evaluation Report

Project evaluation notes

June 2026

Abstract

This report reviews the current `real_attack_eval` results. The evaluation uses the trained Phase 5 fusion detector and runs it on victim-side syscall traces from live attack scenarios. The detector is not retrained in this folder. It is tested as-is against both ADFA-LD attack families and novel families that were not part of ADFA-LD training. The current results show strong trace-level attack presence detection, but weaker and less stable window-level calibration on several families.

1 Overview

The `real_attack_eval` folder is an evaluation harness, not a training pipeline. Its purpose is to answer a practical question:

If the Phase 5 detector is trained and calibrated on ADFA-LD, how does it behave on newer victim-side traces from live attack runs?

The folder contains a local copy of the inference code, selected model artifacts, a shared ADFA-LD benign window pool, and one subfolder per attack family. Each family stores its own manifest, per-window score file, and metrics file.

The current evaluation covers eleven reported rows. Hydra is one capture family, but its FTP and SSH results are reported separately because they have different syscall profiles.

2 Evaluation Protocol

The active detector is the Phase 5 XGBoost meta-classifier from the 32-bit pipeline. It combines three anomaly scores:

- Path A: TF-IDF + PCA + SGD one-class SVM;
- Path B: CNN-1D autoencoder;
- Path C: GRU next-syscall predictor.

The Phase 5 manifest reports `AUC_test = 0.9113` on the original held-out ADFA-LD protocol. The operating threshold used in this evaluation is

$$\tau^* = 0.401214.$$

All family metrics are computed with the same settings:

Table 1: Families included in the current evaluation.

Family	Group	Scenario count	Capture target
Hydra FTP	ADFA-LD family	1	vsftpd worker under repeated FTP login attempts
Hydra SSH	ADFA-LD family	1	OpenSSH worker under repeated SSH login attempts
Adduser	ADFA-LD family	6	SSH session and child shell running user-addition variants
Web Shell	ADFA-LD family	8	Apache/PHP workers after DVWA upload and web shell execution
Meterpreter	ADFA-LD family	10	vsftpd 2.3.4 backdoor shell workflow
Java Meterpreter	ADFA-LD family	8	Java runtime and child processes for MSF Java Meterpreter payload
Reverse Shell	Novel family	6	Python bind shell and spawned bash process
SQLMap	Novel family	5	Apache/PHP/SQLite handling SQLMap probes
Nikto	Novel family	4	Apache workers under Nikto scan profiles
Shellshock	Novel family	6	Apache CGI path and bash children for CVE-2014-6271 variants
Cryptominer	Novel family	4	Fake miner process with different thread counts

- window size: 20 syscalls;
- stride: 2 syscalls;
- syscall ABI: i386;
- benign pool: 50,000 ADFA-LD benign windows;
- benign test split: 40% by trace group;
- random seed: 42.

The data flow is:

1. capture victim-side syscall traces with `strace -f`;
2. filter traces to the victim process tree;
3. map syscall names to i386 syscall numbers;
4. write ADFA-LD-style space-separated syscall sequences;
5. score sliding windows with the Phase 5 inference stack;
6. compute window-level metrics and trace-level detection rates.

The score rows inspected during this review use the i386 mapping and show `oov_rate = 0.0` in sampled traces. This makes a gross syscall ABI or name-mapping error unlikely. The weaker results are better explained by environment shift and threshold calibration.

3 Main Results

Table 2 is the main result table. It should be read before the larger metric table. It includes only the metrics needed to understand the current state: number of traces, trace-level detection rate, window-level detection rate, AUC, and AUCPR(1:1).

Table 2: Main real-attack evaluation results.

Family	Traces	Trace-DR	Window-DR	AUC	AUCPR(1:1)
Hydra FTP	15,272	100.0%	93.8%	0.8684	0.7945
Hydra SSH	5,360	100.0%	58.6%	0.7451	0.6460
Adduser	3,393	100.0%	58.4%	0.7364	0.6475
Web Shell	2,584	100.0%	85.3%	0.8639	0.7781
Meterpreter	2,000	100.0%	81.5%	0.7925	0.6707
Java Meterpreter	21,536	100.0%*	30.1%	0.4723	0.4773
Reverse Shell	1,331	100.0%	66.2%	0.7289	0.5968
SQLMap	1,801	100.0%	98.6%	0.8900	0.7986
Nikto	1,103	100.0%	99.2%	0.8608	0.7668
Shellshock	1,083	100.0%	29.5%	0.6227	0.5505
Cryptominer	1,203	100.0%	55.3%	0.7040	0.5981

*Java Meterpreter detects 21,535 of 21,536 traces; rounded Trace-DR is still 100.0%.

Across all families, the artifacts contain 56,666 attack traces. The trace-level rule detects 56,665 of them. Across windows, 8,268,189 of 13,138,516 attack windows are above the fixed threshold, giving an aggregate Window-DR of 62.9%.

Table 3: Full window-level metrics from the current artifacts.

Family	Attack windows	Benign test	AUC	AUCPR	AUCPR(1:1)	F1@tau	TPR@1% FPR	TPR@0.1% FPR	Median	P10
Hydra FTP	358,565	19,480	0.8684	0.9855	0.7945	0.9606	0.0638	0.0000	1.0000	0.7879
Hydra SSH	2,738,916	19,480	0.7451	0.9958	0.6460	0.7377	0.0035	0.0000	0.5789	0.4954
Adduser	724,334	19,480	0.7364	0.9850	0.6475	0.7339	0.0109	0.0000	0.5829	0.2174
Web Shell	53,244	19,480	0.8639	0.9010	0.7781	0.8729	0.0003	0.0000	0.8500	0.8500
Meterpreter	58,264	19,480	0.7925	0.8508	0.6707	0.8548	0.0000	0.0000	0.8333	0.7143
Java Meterpreter	3,853,379	19,480	0.4723	0.9936	0.4773	0.4626	0.0000	0.0000	0.5732	0.0667
Reverse Shell	1,849,192	19,480	0.7289	0.9898	0.5968	0.7952	0.0000	0.0000	0.4706	0.1481
SQLMap	752,669	19,480	0.8900	0.9930	0.7986	0.9893	0.0047	0.0000	0.9921	0.9632
Nikto	2,677,437	19,480	0.8608	0.9976	0.7668	0.9951	0.0215	0.0000	0.9937	0.9887
Shellshock	44,799	19,480	0.6227	0.7347	0.5505	0.4168	0.0216	0.0000	0.2955	0.1579
Cryptominer	27,717	19,480	0.7040	0.6755	0.5981	0.6334	0.0000	0.0000	0.5714	0.4706

Raw AUCPR and F1@tau are kept in Table 3 because the artifact files store them. They are useful for auditing a run, but they are not the best cross-family comparison metrics. AUC, AUCPR(1:1), Window-DR, and low-FPR TPR are easier to compare across families.

4 Comparison with ADFA-LD Baselines

The ADFA-LD-family rows have paper or LOAO reference values. Table 4 shows the gap between those reference values and the current real captures.

Table 4: ADFA-LD-family comparison. Delta is Real AUC minus Paper AUC.

Family	Paper AUC	Real AUC	Delta AUC	Paper AUCPR(1:1)	Real AUCPR(1:1)
Hydra FTP	0.9112	0.8684	-0.0428	0.8786	0.7945
Hydra SSH	0.8970	0.7451	-0.1519	0.8694	0.6460
Adduser	0.9052	0.7364	-0.1688	0.8676	0.6475
Web Shell	0.9073	0.8639	-0.0434	0.8814	0.7781
Meterpreter	0.9140	0.7925	-0.1215	0.8933	0.6707
Java Meterpreter	0.9140	0.4723	-0.4417	0.8742	0.4773

Hydra FTP and Web Shell transfer better than the other ADFA-LD families. Their real traces still contain dense bursts of I/O, fork, and network activity that resemble the attack signal learned from ADFA-LD.

Hydra SSH, Adduser, Meterpreter, and Java Meterpreter show larger gaps. The largest gap is Java Meterpreter. Its JVM runtime adds many windows dominated by synchronization, memory mapping, and library behavior. Those windows dilute the payload signal and make the window ranking weak.

5 Novel-Family Zero-Shot Results

The novel families were not part of ADFA-LD training. They do not have paper baseline constants. The goal here is not to measure a gap against ADFA-LD paper rows, but to check whether the detector fires on attack behavior it did not see as a named training family.

Table 5: Novel family descriptions and expected syscall signals.

Family	Why it is novel here	Expected syscall signal	AUC	Window-DR
Reverse Shell	Bash-native socket-to-shell control path	socket, exec, read, and write bursts mixed with shell activity	0.7289	66.2%
SQLMap	SQL injection tool, not an ADFA-LD family	repeated HTTP requests, PHP handling, and SQLite file I/O	0.8900	98.6%
Nikto	Web vulnerability scanner outside ADFA-LD families	dense request and response loops across Apache workers	0.8608	99.2%
Shellshock	CGI/bash CVE workflow absent from ADFA-LD	short bash execution phase after Apache CGI handling	0.6227	29.5%
Cryptominer	CPU resource-abuse simulation absent from ADFA-LD	thread and CPU-loop behavior with limited syscall diversity	0.7040	55.3%

SQLMap and Nikto transfer best. They produce repeated web probing behavior, so the abnormal pattern appears in most windows. Reverse Shell and Cryptominer are in the middle. Shellshock

is the weakest novel row: every trace is detected at least once, but most windows remain below threshold.

6 Interpretation

6.1 Trace-level detection is strong

The trace-level rule is sensitive. Almost every attack trace has at least one window above τ^* . This supports a limited but useful claim: the detector often recognizes attack presence in real traces, including in families that were not part of ADFA-LD training.

6.2 Trace-DR is not the same as deployment quality

Trace-DR can be high even when the window ranking is weak. A trace is counted as detected after one flagged window. Long traces have many chances to cross the threshold. For that reason, Trace-DR should always be shown with Window-DR, AUC, and low-FPR TPR.

Java Meterpreter and Shellshock make this point clearly. Both are detected at trace level, but their Window-DR values are only 30.1% and 29.5%. That means the detector sees some abnormal windows, not that every phase of the attack is cleanly localized.

6.3 The main weakness is calibration under environment shift

The current threshold was calibrated on ADFA-LD. The real traces come from newer services, newer runtimes, and a newer kernel. Modern OpenSSH, Apache/PHP, JVM, and system utilities emit syscall patterns that can look benign relative to the old ADFA-LD distribution.

This explains the shape of the results. SQLMap and Nikto remain strong because their attack behavior is dense and repeated. Java Meterpreter weakens because the JVM runtime dominates many windows. Shellshock weakens because the exploit phase is short.

6.4 No obvious ABI failure is visible

The sampled score rows use the i386 syscall ABI and show no OOV mapping problem. This does not prove preprocessing is perfect, but it argues against a simple syscall-name mapping bug as the main reason for weaker rows.

7 Conclusion

The current real-attack evaluation supports a careful conclusion. The Phase 5 fusion detector generalizes well enough to flag nearly all real attack traces at least once, including several novel families. That is an attack-presence result.

The stronger operational claim is not yet supported. Window-level ranking and low-FPR behavior vary across families, especially Java Meterpreter, Shellshock, Hydra SSH, and Adduser. The likely cause is the combination of kernel/runtime shift, service overhead, and a fixed threshold calibrated on ADFA-LD.

For reporting, the safest wording is:

The detector shows strong trace-level attack-presence detection on real captures, but its ADFA-LD-calibrated threshold does not provide stable low-FPR window-level behavior across all families.